



## Technical Reproducibility Validation Report

### STRsearch

Version c70179b

Autosomal STR

|                   |                                                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Verification date | 2026-09-14                                                                                                                                 |
| Repository commit | <a href="https://github.com/STRhub/STRsearch/commit/c70179b3b175adc82a7314409af06900b3861d61">c70179b3b175adc82a7314409af06900b3861d61</a> |
| Attestation level | <b>Runs + Plausible output</b>                                                                                                             |
| Permalink         | <a href="https://strhub.app/verified/strsearch-c70179b">https://strhub.app/verified/strsearch-c70179b</a>                                  |
| STRhub            | <a href="https://strhub.app">https://strhub.app</a>                                                                                        |

#### SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

#### WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

#### NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance



## 1. Executive Summary

|                   |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| PURPOSE           | Verify that the tool installs, runs end-to-end, and produces structurally valid output. |
| RESULT            | <b>5/5 gates passed. Runs + Plausible output</b>                                        |
| ATTESTATION LEVEL | <b>Runs + Plausible output</b>                                                          |
| SCOPE             | Installation, execution, and output structure verification.                             |
| NOT EVALUATED     | Genotype accuracy · Concordance · Forensic validity · Regulatory compliance             |

## 2. Reproducibility Metadata

|                 |                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tool            | STRsearch                                                                                                                                             |
| Version         | c70179b                                                                                                                                               |
| Repository      | <a href="https://github.com/AnJingwd/STRsearch">https://github.com/AnJingwd/STRsearch</a>                                                             |
| Commit (pinned) | <a href="https://github.com/AnJingwd/STRsearch/commit/c70179b3b175adc82a7314409af06900b3861d61">c70179b3b175adc82a7314409af06900b3861d61</a>          |
| Container OS    | ubuntu-22.04                                                                                                                                          |
| Marker panel    | Autosomal STR                                                                                                                                         |
| Verified on     | 2026-09-14 20:32:41 UTC                                                                                                                               |
| CI run          | <a href="https://github.com/Tfronta/strhub-verified/actions/runs/34893427310">https://github.com/Tfronta/strhub-verified/actions/runs/34893427310</a> |
| Submitted by    | A third party (not the tool's maintainer)                                                                                                             |

This tool was submitted for verification by somebody other than its maintainer. The maintainer took no part in the run and supplied none of what it used: the command, the environment, and any target regions were chosen by the submitter. This certificate records a run; it is not an endorsement by the tool's maintainer, and does not imply their involvement.

## 3. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRsearch c70179b · commit c70179b
$ python3 /opt/STRsearch/pipeline.py \
--type paired \
--num_processors 4 \
--reads_threshold 30 \
--stutter_ratio 0.5 \
from_bam \
--working_path /data/out \
--sample sample \
--sex female \
--bam /data/in/input.bam \
--ref_bed /data/in/regions.bed \
--genotypes /data/out/genotypes.txt \
--multiple_alleles /data/out/multiple_alleles.txt \
--qc_matrix /data/out/qc_matrix.txt
```

## 4. Verification Gates

|                  |                                                           |      |
|------------------|-----------------------------------------------------------|------|
| Available        | The pinned public source exists                           | PASS |
| Installs         | The environment builds from source                        | PASS |
| Runs             | Executes end-to-end without crashing                      | PASS |
| Expected IO      | Produces a non-empty file in the declared format          | PASS |
| Plausible Output | Output structure matches expected genotype-bearing format | PASS |
| Execution log    | <a href="#">strsearch-c70179b.log-external.txt</a>        |      |

## 5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

## 6. Output Content Evidence

|                          |                      |
|--------------------------|----------------------|
| Output file              | multiple_alleles.txt |
| Format                   | TSV                  |
| Sequence records         | 388                  |
| Distinct STR loci        | 24                   |
| Total reads              | 2,401                |
| Max depth (single locus) | 51                   |

## 7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

|             |                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset     | Illumina BAM (hg38), NA12878 (autosomal, female)                                                                                                                                                                       |
| Source      | <a href="https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat...">https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat...</a>                                                                          |
| License     | Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.                                                                                                                              |
| Ref. genome | GRCh38 / hg38                                                                                                                                                                                                          |
| Loci tested | CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · SE33 · TH01 · TPOX · vWA |
| Regions BED | Provided by STRhub                                                                                                                                                                                                     |

## 8. Verification Matrix

|             |                             |                                                  |
|-------------|-----------------------------|--------------------------------------------------|
| <b>PASS</b> | Reference dataset           | Illumina BAM (hg38), NA12878 (autosomal, female) |
|             | README check (5/5)          | Advisory: all items present                      |
|             | Install / environment setup | Present                                          |
|             | Run command                 | Present                                          |
|             | Expected input format       | Present                                          |
|             | Produced output format      | Present                                          |
|             | Dependencies / versions     | Present                                          |

## 9. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

- Test data: no sample from the repository was used, so a public reference sample stood in.
- A container environment, supplied with the submission.

## 10. Limitations

- Single reference dataset per input type
- Single containerized environment (Docker / ubuntu-22.04)
- No truth-set comparison or ground-truth genotypes
- No accuracy or concordance assessment
- No forensic validation of results
- Short-read limitations apply (very long STR alleles may not span reads)

## 11. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

## 12. Conclusion

### ◆ Runs end-to-end

STRsearch c70179b installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

### ◆ Produces valid output

The tool generated a structurally valid TSV output file with genotype calls across 24 target forensic STR loci, with a maximum read depth of 51 at vWA.

### ◆ No bundled demo data

STRsearch does not include its own test or demo dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

### ◆ Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.



| LOCUS    | READ DEPTH |                      | LOCUS    | READ DEPTH |        |
|----------|------------|----------------------|----------|------------|--------|
| vWA      | 241        | ████████████████████ | CSF1PO   | 92         | ██████ |
| D21S11   | 187        | ██████████████████   | D6S1043  | 92         | ██████ |
| FGA      | 164        | ██████████           | D2S441   | 88         | █████  |
| D19S433  | 131        | ██████████           | D1S1656  | 81         | █████  |
| SE33     | 125        | ██████████           | PentaE   | 78         | █████  |
| D18S51   | 123        | ██████████           | D13S317  | 66         | ████   |
| D16S539  | 105        | ██████               | D10S1248 | 65         | ████   |
| D3S1358  | 99         | ██████               | D2S1338  | 62         | ████   |
| PentaD   | 99         | ██████               | D8S1179  | 62         | ████   |
| D7S820   | 96         | ██████               | TPOX     | 55         | ███    |
| D22S1045 | 95         | ██████               | D12S391  | 55         | ███    |
| D5S818   | 94         | ██████               | TH01     | 46         | ███    |